Mahmoud Alsharnouby

Mechatronics Engineer

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Mechatronics & Robotics Engineer with 8+ years of experience in mechanical design, sub-assembly modeling, prototyping, and design validation across manufacturing and engineering services sectors. Proficient in SolidWorks (CSWP), AutoCAD, ANSYS, and GD&T. Strong focus on design-for-manufacturability, risk identification, and Quality Management Systems (QMS) including ISO 13485 for medical device design and development. Data analytics background (Python, Power BI) enables data-informed design decisions and cross-functional collaboration with electronics and manufacturing teams. Currently based in Lisbon, Portugal, and open to relocate worldwide for the right opportunity.

EXPERIENCES:

Engineering Consultant - Mechanical & Industrial · Meta Plat Inc. — March 2021 - Apr 2026

- Provided mechanical engineering consultancy to industrial clients, supporting design workflow optimization and engineering KPI performance across manufacturing projects.
- Developed engineering performance monitoring tools using Power BI and Tableau to track mechanical asset health and production metrics in real time.
- Utilized Python (Pandas, NumPy) and SQL to analyze engineering datasets, enabling data-driven mechanical design decisions in industrial environments.
- Advised manufacturing engineering teams on integrating IoT and Digital Twin solutions into mechanical systems for predictive maintenance and industrial automation.
- Collaborated with mechanical engineering teams to streamline CAD design workflows and product lifecycle management strategies across client projects

Mechanical Design and Digital Twin Engineer - PMICO Inc. — May 2018 - Nov 2020

- Designed advanced mechanical systems and steel structures for oil & gas, power generation, and manufacturing projects.
- Built digital twin models using SolidWorks and ANSYS for real-time asset monitoring and predictive maintenance.
- Applied DFM principles and sub-assembly risk identification to optimize designs and reduce manufacturing costs.
- Collaborated with cross-functional teams to deliver data-driven engineering solutions for multinational clients.

Mechanical Design and Systems Integration Engineer - KONE EG — May 2015 - Apr 2018

- Designed mechanical systems for elevators and escalators, integrating with electrical, software, and BMS platforms.
- Conducted V&V activities for mechanical sub-assemblies, ensuring compliance with EN 81, EN 115, and KONE standards.
- Utilized SolidWorks, AutoCAD, and Revit to produce detailed 3D designs and simulations for multiple client projects.
- Collaborated with multidisciplinary teams to reduce installation time and improve system reliability.
- Delivered training on systems integration best practices to cross-functional teams across multiple projects.

SELECTED PROJECTS:

- Safety Impact Structure - Toyota Fortuner | Schlumberger Full design lifecycle: concept → 3D modeling → FEA validation (1,736 kN at 150 km/h) → GD&T drawings → fabrication support. Applied DFM and risk mitigation principles throughout.

- Offshore Spool Winch - Client |Halliburton
  Hydraulic offshore winch designed to DNV standards for umbilical deployment.

- Offshore Site Container - Client |Weatherford International
  Designed to EN 12079-2006 and DNV 2.7-1 standards.

- Mobile Shale Shaker System - Client: Egyptian Drilling Company (EDC)
  Mobile mud system with 8 hydraulic cylinders for oil & gas drilling operations.

- Elevator & Escalator Upgrade - Cairo Metro Line 3 Client |KONE EG / Cairo Metro Authority
  Modified and upgraded elevator and escalator mechanical systems for Cairo Metro Line 3 stations, compliant with EN 81, EN 115, and KONE global standards.

EDUCATION:

Master of Sc. Engineering Major Mechatronics, - University of Evora, PT. 2027

Bachelor of Sc. Major Mechatronics and Robotics, - University of Wales, UK. 2015.

SKILLS:

Mechanical Design Engineer specializing in Digital Twin development and IoT systems integration, with strong analytical and data visualization capabilities.

Technical Skills

Engineering & Design Software
- CAD/Simulation: SolidWorks (CSWP), AutoCAD, Siemens NX, ANSYS, Revit.
- Prototyping: LabVIEW, Proteus

IoT & Connected Systems (In Progress)
- Currently studying: MQTT, AWS IoT Core

Digital Twin & Predictive Maintenance
- Real-time simulation and monitoring
- AI/ML integration for predictive maintenance

Data Analysis & Visualization
- Tools: Excel (Advanced), Tableau, Power BI, SQL
- Languages: Python (Pandas, NumPy), DAX
- Methods: Data cleaning (Excel, Power Query)

Programming & Development
- Languages: Python, Kotlin, Embedded C (concepts), MATLAB and Simulink.

Languages

English (Fluent)
Arabic (Native)
Portuguese (Intermediate)

Interpersonal Skills

- Multilingual communicator (English, Arabic, Portuguese)
- Cross-cultural team collaboration experience
- Fast learner with proven adaptability in international environments

CERTIFICATES:

- Electrified Systems Design Engineer Professional Certificate
- Self-Driving Cars Specialization
- ISO 13485-Medical Device Design & Development Control
- GD&T: Advanced Concepts
- FEA: Introduction to Finite Element Analysis
- SolidWorks Certified Professional (CSWP)
- Lean Six Sigma Green Belt (6σ)
- Internet of Things Business Impact
- ROV Technical Certificate
- Robotics Software Engineer Nanodegree
- CNC Programming & Coding Systems Certificate
- Machine Design Series Certificate
- Google Data Analytics Professional Certificate

Social duties

Awards:
- 1st Place - Future City Competition, IEEE (2014)

Volunteering:
- Web Summit Lisbon — Tech Volunteer | 2023 – Present
Engaged with global IoT, AI & startup ecosystem at one of the world's largest tech conferences.
- Portuguese Blood Bank — Volunteer | 2024 – Present
Supporting community health initiatives in Lisbon.